

Adding a Node to an Azure Local Cluster



Agenda

- **Objective:** Step through the process of adding a node to an Azure Local cluster
- **Scope:** Capabilities and constraints, hardware and software prerequisites, implementation workflow, monitoring and troubleshooting.



Scaling Azure Local Capacity (Hyperconverged)

- Scale compute and storage capacity simultaneously
- Supported cluster size: up to 16 nodes
- Only one node can be added at a time
- Multiple nodes can be added sequentially
- Orchestrator (LCM) automatically updates resiliency, storage, and networking
- Storage rebalance occurs automatically after each addition
- Node removal is not supported

Supported Scale-Out Scenarios

- 1 → 2 nodes: two-way mirror, witness required
- 2 → 3 nodes: three-way mirror, switch required
- 3 → N nodes: three-way mirror, switch required
- Witness optional (except 1→2 transition)

Hardware Requirements

- Node must match CPU, memory & drive count/type
- Must use same OEM hardware & validated components
- CPU/memory mismatches generate warnings (not blocking)
- Drive count mismatches block the operation
- Hardware properly racked & cabled
- Switch ports enabled/configured

Software Requirements

- Active Directory user AzureStackLCMUser active
- Admin must sign in using LCM user account or equivalent
- Arc registration uses same parameters as existing nodes

Preparing the New Node

- Install Azure Local OS (matching solution version)
- Drivers & firmware updated
- Register node with Azure Arc
- **Assign required RBAC roles to the node**
 - Azure Stack HCI Device Management Role and Key Vault Secrets User
 - Allow the orchestrator to retrieve secrets and manage the node throughout its lifecycle

Running Add-Server

- Run `Add-Server` with name, IPv4, and credentials
- System validates CPU/memory/drive requirements
- After success, storage pool automatically rebalances
- Custom storage IPs require manual reassignment

Monitoring & Troubleshooting

- Monitor operation using:
 - `Start-MonitoringActionplanInstanceToComplete`
 - `Get-VirtualDisk` and `Get-StorageJob` for rebalancing
- Sync node visibility with: `Sync-AzureStackHCI`
- Use logs & `Add-Server -Rerun` for failures

Recovery Scenarios

- Out-of-band node addition:
 - Unsupported → remove node
- Failed orchestrator operation:
 - Rerun **Add-Server**
- Partial success requiring OS reinstall:
 - Use repair workflow
- Collect logs with **Send-DiagnosticData**

cloudops *gility*